



Snipe-IT

Asset Management

Software

Project Documentation.

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Introduction.

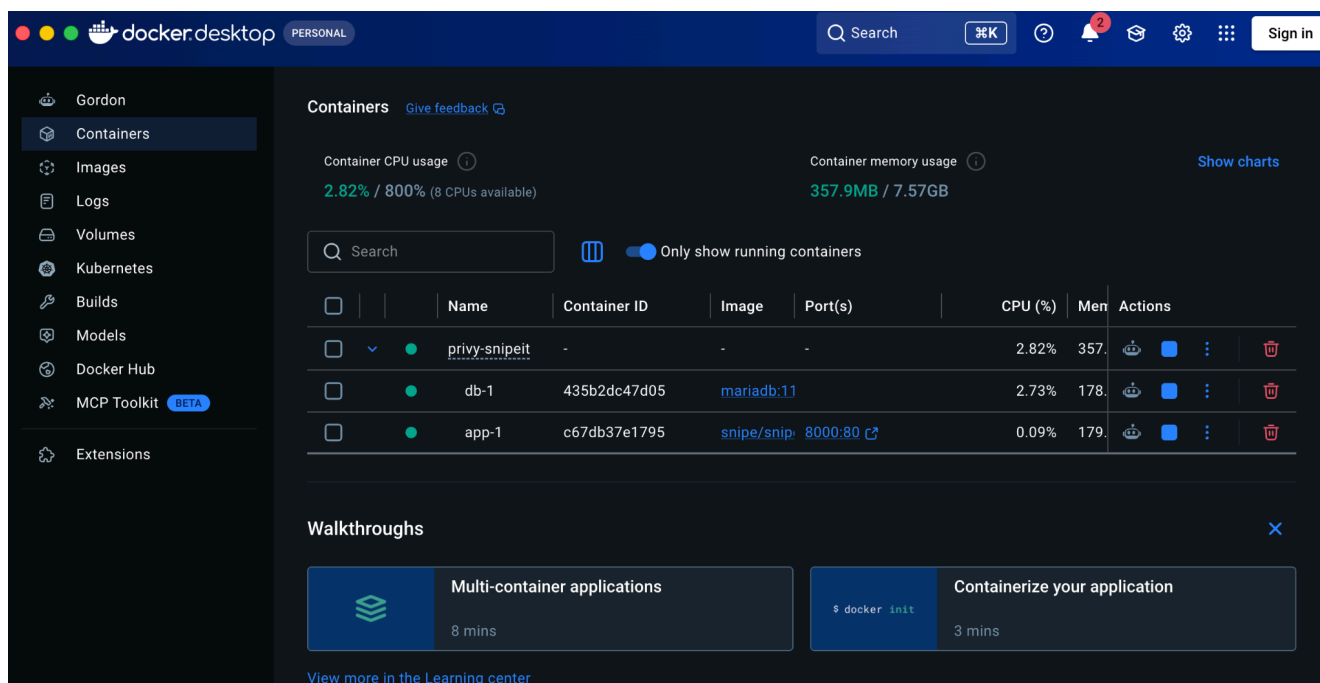
Snipe-IT is an open-source IT asset management software which allows users to quickly and easily manage licenses, devices, and peripherals inside an organisation. This project was set up to manage all of Privy's assets.

Infrastructure.

The backend infrastructure consists of the Docker deployment of the server (including environment set-up), and GitHub repo for replication. These services work in tandem to ensure uptime, transparency, and to maintain collaboration between project members.

Docker Containers

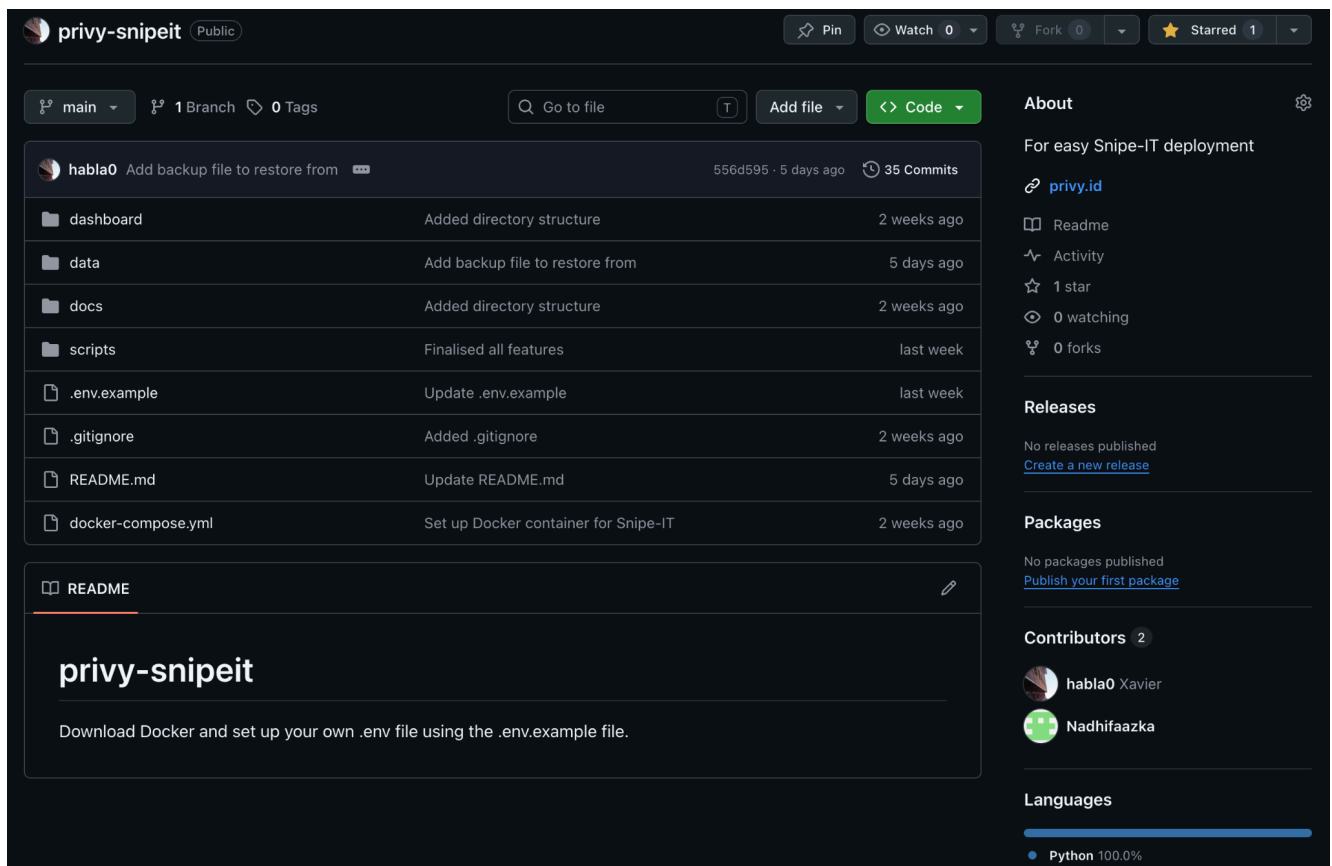
Snipe-IT runs inside a Docker container which handles the database and server app. It is deployed from the official Snipe-IT image, running Snipe-IT version 8.6.2. It can be started using the Desktop App or through the terminal by calling `docker compose up -d`.



Using a Docker container ensures the server can run locally across all machines when shared across GitHub, and it simplifies deployment. Only the .env file needs to be edited to include the user's API and app keys. Changes from the default settings in the .env setup include the addition of an API key, and the increase of the API request limit.

GitHub Repository

A GitHub repository was also set-up to facilitate real-time file sharing, conflict management, and source control. Every change made to the files in the repository can be tracked transparently and easily — and scaling up deployment is as easy as sharing the repository link. The repository is also particularly helpful for managing script and csv versions and any changes can easily be rolled back. This provides more control but less automation compared to a cloud file service like Microsoft OneDrive or Google Drive.

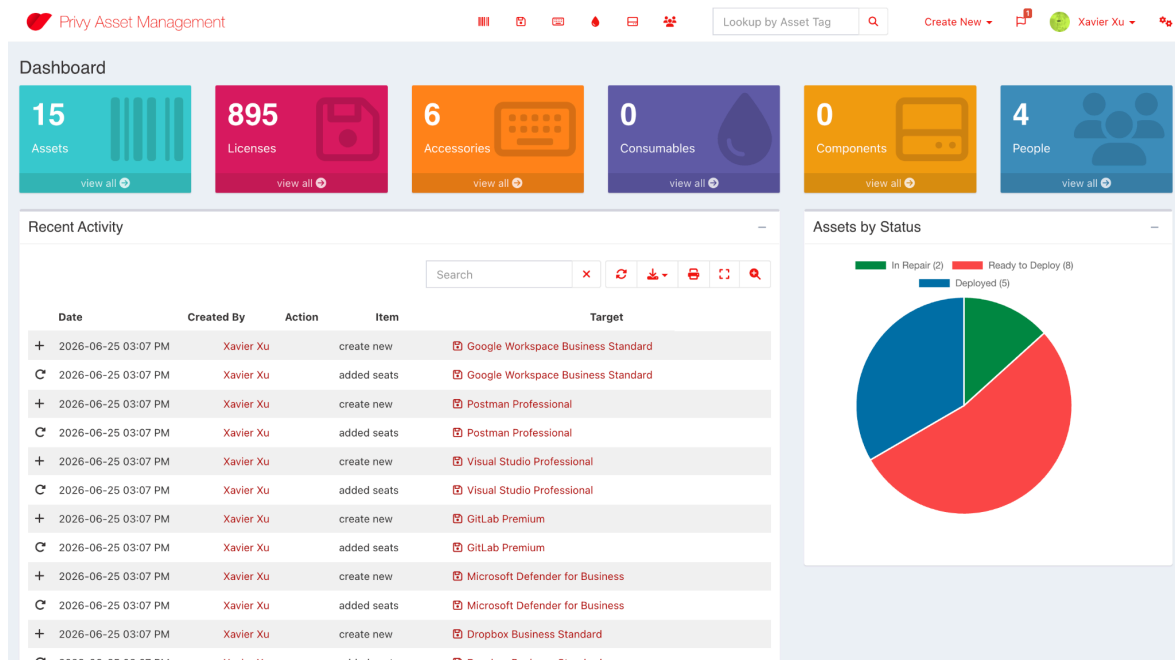


The Application Layer.

The main application layer is a customised Snipe-IT deployment specialised towards handling the type of corporate assets and accessories (laptops, monitors, mice, etc.) that would be present in an environment like Privy. The application allows you to add, remove, and set attributes for objects (assets, licences, users, etc.) inside the interface or import them via a CSV, as well as featuring a complete API to automate the addition and removal of objects.

As Snipe-IT is open-source software, the documentation on specific Snipe-IT features is available publicly on their website and anybody can contribute to adding features to the main application. This set of documentation relates only to Privy's implementation of Snipe-IT.

Basic Usage Information



The Dashboard is the central hub where objects can be managed and recent activity from users and admins tracked. The system is set up with a general "superadmin" account with Xavier's account also as a superadmin. Every asset is tracked and graphed based on availability. Asset imports and some asset management can also be achieved through a script, although for consistency, it is highly recommended to keep management inside the app.

Installation.

1. Installing Git (or GitHub Desktop)
 - a. Install Git from the website or GitHub Desktop, and follow the instructions
 - b. Download the repository from GitHub (<https://github.com/habla0/privy-snipeit>) onto your local device
2. Installing Docker Desktop
 - a. Download and install the Docker Desktop app
3. Install Python 3.14 or above
4. Setting up the .env
 - a. Copy the .env.example file and rename the copy to .env
 - b. Follow the steps in the file to add the required information
5. Start the Docker container
6. Run through the setup on localhost:8000
7. Import data via the automation script

Data Model.

The asset management system is implemented using Snipe-IT to manage and track IT assets across multiple categories, such as laptops, desktops, phones, monitors, desktops, keyboards, mice, and software. To ensure data consistency, a structured data model was created consisting of Custom Fields, Status Labels, Categories, Manufacturers, and Asset Models.

Model Definitions

Data Dictionary

Field	Description	Example
Asset Tag	Unique identifier for each asset	A0001
Model	Asset model name	Latitude 5440
Category	Asset type	Laptop
Manufacturer	Asset brand	Dell
Status	Current asset status	Ready to Deploy, Deployed, In Repair, Lost, Archived
Serial	Serial number	DLL1111
Mac Address	Unique network interface identifier	3C:52:82:A1:B2:C1
CPU	Processor specification of the asset	Intel Core i5-1335U
RAM	Memory capacity of the asset	16 GB
Storage	Storage capacity of the asset	512 GB SSD
Operating System	Installed operating system	Windows 11 Pro
Graphics	Graphics processing unit specification	Intel Iris Xe Graphics
Purchase Date	Asset purchase date	2024-01-03 (YYYY-MM-DD)
Location	Physical asset location	Privy HQ
Name	Name of the software license	Microsoft 365 Business Standard
Seats	Total number of available license seats	100
category_id	Reference ID of the license category	1

Custom Fields

Additional attributes created for store asset information that is not provided by Snipe-IT default fields. By using custom fields, asset information can be managed more detailed, structured, and consistent. Custom fields are grouped into a fieldset to ensure consistency in asset data.

- MAC Address
- CPU
- RAM
- Storage
- Operating System
- Graphics
- Warranty Expiry

Status Labels

Status labels are created to indicate the current condition and lifecycle stage of all assets within the inventory systems. Status Labels help track asset availability, usage, maintenance and retirement throughout the asset lifecycle.

- In Repair – Asset is temporarily unavailable due to under maintenance or in repair
- Ready to Deploy – Asset is available and ready to be assigned to users
- Archived – Asset is retired and no longer in active use
- Lost – Assets are lost and their whereabouts are unknown
- Deployed – Asset is being actively used and has been assigned to users

Categories

Categories are used to classify assets based on their type and function, making asset management more structured.

- Misc Software
- Laptop
- Phone
- Monitor
- Desktop
- Keyboard
- Mouse

Manufacturers

Manufacturers represent the companies or brands that produce assets, enabling assets to be easily identified based on their manufacturer or brand.

- Acer
- Apple
- Dell
- HP

- Lenovo
- Logitech
- Microsoft
- Samsung

Location

Locations represent the physical locations where assets are stored or used, enabling effective asset tracking across offices and facilities.

- Privy HQ
- Privy Quarter

Usage Workflow.

Asset Management

Asset Request Process

End User:

1. Request asset to the admin
2. The request is reviewed by the admin
3. If an asset is available, it will be assigned to the user

Admin:

1. Review the asset request
2. Check asset availability in Snipe-IT
3. Select an asset with the status "Ready to Deploy"
4. Check out the asset to the user
5. Confirm that the asset status changes to "Deployed"

Asset Checkout

Asset checkout is performed when an asset is assigned to a user.

Steps:

1. Open the asset list
2. Choose the asset and click "Checkout"
3. Select user
4. Click "Checkout" button

Results:

- Asset is assigned to the selected user
- Asset status change to "Deployed"

Asset Check-in

Asset check-in is performed when an asset is returned.

Steps:

1. Open the asset list
2. Choose the asset and click "Checkin"
3. Click "Checkin" button

Results:

- Asset is no longer assigned to a user
- Asset status change to "Ready to Deploy"

Asset Repair

Repair is performed when an asset requires maintenance or repair.

Steps:

1. Open the asset list
2. Choose the asset and click the pencil icon for update
3. Change the status to "In Repair"
4. Save the changes

Result:

- Asset is temporarily unavailable

Asset Archive

Archive is performed when an asset is retired or no longer is used.

Steps:

1. Open the asset list
2. Choose the asset and click the pencil icon for update
3. Change the status to "Archived"
4. Save the changes

Result:

- The asset is removed from active use and can't be assigned to users.

CSV Importing

The CSV import feature allows admin to upload multiple asset records into Snipe-IT at once.

Steps:

1. Prepare the asset inventory data in CSV format
2. Choose "Import" menu in Snipe-IT

3. Upload the CSV file
4. Map each CSV column to the corresponding Snipe-IT field
5. Review the field mapping to ensure all fields are correctly assigned
6. Click "Import" process
7. Review the import result
8. If the import result error or fail to import, fix the error and repeat the import process (from step 1)
9. Ensure imported assets contain the correct category, status, and custom field information

Results:

- Asset inventory data is successfully imported into Snipe-IT
- Asset records are stored with the correct categories, statuses, and custom field values
- Multiple asset records can be uploaded at once, reducing manual data entry and improving efficiency